Ali Rıza Ercan

Data Scientist & Software Developer

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PROFILE

Data Scientist and Software Developer working as an R&D engineer at Autodidactic Technologies, building real-time industrial quality analytics end to end and researching LLM-guided reinforcement learning in a TÜBİTAK-funded project. Computer Engineering graduate (GPA 3.78/4.00) with international experience in Poland and a TEKNOFEST 2024 runner-up.

EXPERIENCE

R&D Engineer · Data Science & Software · Autodidactic Technologies — Jan 2026 – Present
Konya, Türkiye

- Build a production platform that turns live sensor data from manufacturing lines into quality insight: real-time ingestion, time-series storage on PostgreSQL/TimescaleDB, batch processing and analysis services in FastAPI, and interactive measurement dashboards in Vue 3.
- Designed scalable analytics on TimescaleDB continuous aggregates and secured the interfaces with Keycloak role-based access; 325 commits and 115 merged pull requests in the last 12 months.
- TÜBİTAK-funded R&D: LLM-generated reward functions for simulated agents (Qwen 2.5 via Ollama), domain fine-tuning of Llama 3 and Qwen 2.5 with Unsloth, and a dogfight sandbox simulation environment.

Data Science & Software Developer Intern · Mind in Blue — Jun 2025 – Jan 2026
Gdańsk, Poland

- Built a responsive, multilingual React web application with RESTful APIs, improving engagement and site performance.
- Automated data pipelines and applied statistical and predictive models to marketing KPIs for data-driven decisions.

Data Scientist Intern · HAVELSAN — Jul 2024 – Aug 2024
Ankara, Türkiye

- Benchmarked Turkish NER and NLP models with scikit-learn and Hugging Face Transformers; the report guided the next project phases.
- Developed a FastAPI microservice and Python dashboards for text preprocessing, model serving and visualisation.

Assistant Student · KTO Karatay University · Technology Transfer Office — Dec 2023 – Jun 2024
Konya, Türkiye

- Analysed and reported patent and research-project data with Excel and Python to support strategic decisions.

Software Developer Intern · MetalMaker 3D — Jul 2023 – Oct 2023
Kraków, Poland

- Developed Python GTK applications on Raspberry Pi for manufacturing control, working with UI/UX designers in Figma and Glade.
- Refactored React and Supabase modules to streamline data flow and enable scalable deployment.

EDUCATION

KTO Karatay University · B.Sc. Computer Engineering · GPA 3.78/4.00 — Sep 2020 – Jun 2025
Konya, Türkiye

Politechnika Bydgoska · Erasmus+ exchange, Computer Science · GPA 3.96/4.00 — Oct 2022 – Mar 2023
Bydgoszcz, Poland

AI and Technology Academy · Data Science Scholar — Jan 2025 – Aug 2025
Online

SELECTED PROJECTS

TYFOR — Football analytics platform with 50+ clubs and 1,500+ player profiles; computer-vision (YOLOv8) scouting network; React, Flask, PostgreSQL, Docker.

No One Waits — Real-time public-transport occupancy prediction with computer vision; TEKNOFEST 2024, 2nd place.

Miss Minutes — Voice-first personal command center in Next.js: LLM tool use, streaming TTS, Google OAuth integrations, listening analytics.

Salifort Motors HR Attrition — Random Forest model with 97.8% accuracy identifying key retention drivers.

Store Sales Forecasting — LightGBM forecasting over 3M+ rows for 54 stores (0.4725 RMSLE).

PassoAssist — BERT-based complaint classification on scraped customer feedback, served with Flask.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C
Data science & AI: scikit-learn, PyTorch, TensorFlow, Hugging Face, LightGBM, YOLOv8, Ollama, Unsloth
Back end & data: FastAPI, Flask, SQLAlchemy, PostgreSQL, TimescaleDB, MariaDB, Redis, BigQuery
Front end: Vue 3, React, Next.js, Konva, Tailwind
Tools: Docker, Keycloak, Git, Linux, Google Cloud, Vercel

AWARDS & CERTIFICATES

- TEKNOFEST 2024 Smart Transportation, 2nd place (No One Waits)
- TEKNOFEST 2023, 4th place (No Standing Passengers)
- Google Advanced Data Analytics Certificate (Coursera, 2025)
- Kaggle Expert · GDSC KTO Karatay Technology Core Team Lead (2024)

LANGUAGES

Turkish: Native
English: B2 · fluent working proficiency